

Alveolar Bone Grafting and Fistula Closure

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INTRODUCTION

The advantages of alveolar bone grafting include bony support for the teeth adjacent to the cleft, enhancement of orthodontic and prosthodontic treatment, stabilization of the maxillary arch, and improvement in oral hygiene through the elimination of oronasal fistulae.¹ Previous techniques such as those by Boyne, Abyholm, and Hall have been described.²⁻⁴ The purpose of this article is to demonstrate the senior author's (DMS) preferred technique, which involves a maxillary vestibular approach, incisions for repair of an anterior sulcus fistula, wide defect exposure, cadaveric bone graft augmentation, and tri-laminar closure. (Fig. 1) (See Video [online], which displays alveolar bone grafting operative technique.)

OPERATIVE TECHNIQUE

The patient described is an 11-year-old girl who presents with an alveolar gap and alveolar sulcus fistula after repair of a left unilateral cleft lip and palate in infancy.

After placement of a Dingman retractor to allow visualization of the intraoral cavity, the intraoral procedure begins with creation of a keyhole incision around the fistula. Buccal flaps on either side of the cleft are raised in a subperiosteal fashion using incisions made bilaterally along the buccal/labial dento-gingivo junction that are continuous with the key-hole fistula incisions. Bilateral relaxing incisions aimed vertically near the first molars are made followed by subperiosteal exposure of the anterior maxilla. The underlying periosteum of the raised buccal flaps is then scored to facilitate later flap mobilization. If the fistula includes a palatal component, the keyhole incision is extended circumferentially along the palatal fistula. Palatal flaps are then raised bilaterally along the lingual dento-gingivo junction and raised in a subperiosteal fashion. At the fistula defect, careful dissection of both the buccal and palatal flaps from each other is performed,

followed by dissection of the nasal floor from these two dissected buccal and palatal layers. Extensive subperiosteal dissection along the piriform aperture and maxillary crest is performed to achieve dissection beyond the cleft defect and fistula in all three layers. The nasal soft tissue is pushed into the nasal cavity and closed in a water tight fashion. After closure of the nasal side, a saline syringe is inserted into one of the nasal passages and egress of saline fluids is evaluated at the nasal closure.

Cancellous bone from the iliac crest is harvested.^{2,5,6} If it appears that there is inadequate autogenous cancellous bone to fill the defect, the bone graft can be augmented with demineralized bone matrix (DBX; Johnson & Johnson) and bone morphogenetic protein to ensure adequate bone mass transplantation. The mixture is tightly packed into the alveolar cavity. Additional bone is placed on the deficient area of the maxilla at the piriform and nasal floor on the cleft side to improve symmetry. The mucoperiosteal buccal and palatal flaps are advanced and closed over the surface of the former fistula/alveolar defect.

CONCLUSIONS

The technique presented here involves a wide defect dissection followed by discrete closures of the trilaminar layers (oral, nasal, and palatal). This creates a large pocket for the bone graft transplantation with a tension-free and watertight closure. Creation of a wide pocket ensures adequate bone stability for future orthognathic surgery.

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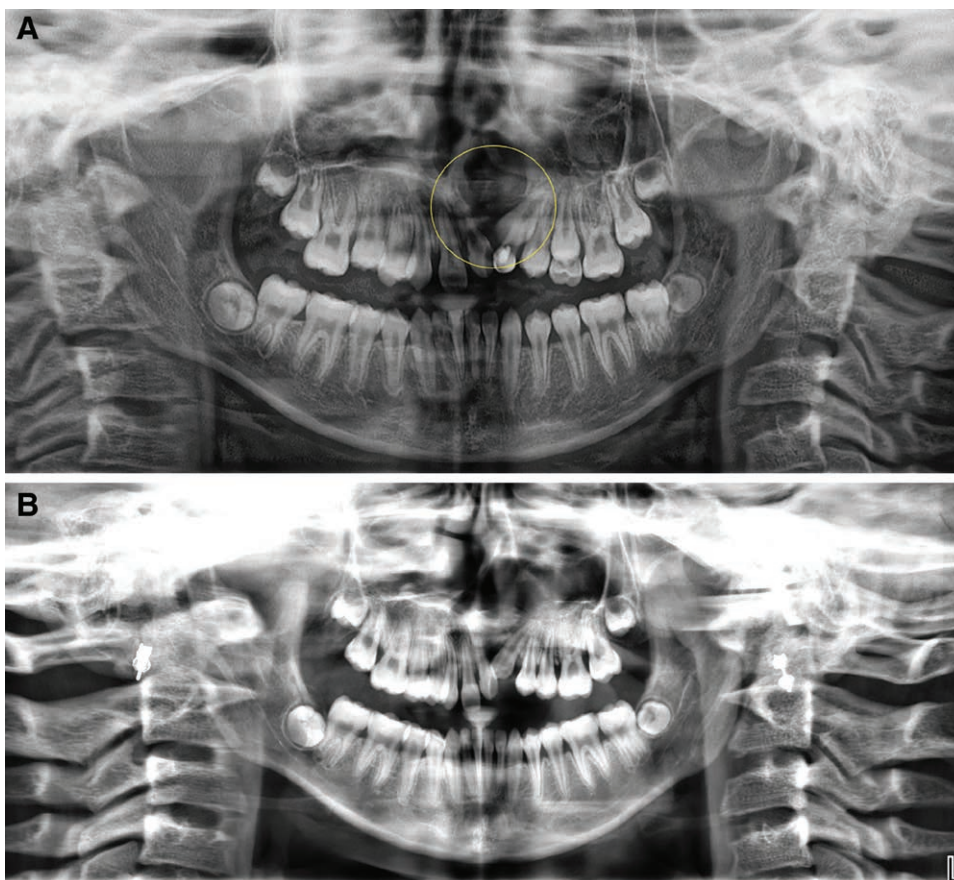


Fig. 1. Panorex demonstrating left unilateral alveolar cleft. Preoperative (A) and postoperative panorex (B) showing bone growth in the left unilateral fistula in the alveolar region in patient with left unilateral cleft lip and palate. Surgery is timed around age 9–12, before the eruption of the permanent canine.

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